

CRR: Canonical Formalism, Processual Axioms, Archetypal Structure, and Platonic Ingression

A working mathematical and comparative synthesis of Coherence-Rupture-Regeneration

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Status: exploratory synthesis; not a peer-reviewed paper.

Scope and provenance

The canonical equations in Sections 1-2 reproduce the current public CRR formulation as presented by temporalgrammar.ai and the CRR repository. The archetypal and Platonic-ingression sections are comparative and constructive extensions developed in the present discussion. They are explicitly marked as interpretations or proposed mathematics rather than established consequences of CRR, the Free Energy Principle, or the cited contemplative traditions.

Abstract

Coherence-Rupture-Regeneration (CRR) proposes a minimal temporal grammar for finite systems: coherence accumulates, a regime closes at rupture, and regeneration reconstructs from causal history. In its current public form CRR is presented as a processual complement to the Free Energy Principle (FEP), with Fisher information geometry providing a candidate endogenous clock and an exponential historical kernel providing non-Markovian reconstruction. This document first sets out the current canonical equations and notation, then distils five processual axioms. It next develops two explicitly comparative extensions. The first reads masculine and feminine archetypes as relational operations - differentiation and holding, emission and reception, line and circle - rather than as fixed traits of men and women. The second treats Platonic or Whiteheadian ingression as a history-conditioned measure over potential forms, coupled to active inference and extended across nested scales. The appendices compare every CRR iteration encountered in the source paper, public website, cross-domain CV paper, canonical repository, and the processual extensions developed here, concluding with a recommendation for a mathematically cleaner full-paper variant.

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1. CRR in brief: a processual clock for finite systems

CRR is best introduced as a grammar of temporal persistence rather than as a claim that one mechanism explains every domain. A finite system must preserve enough structure to remain identifiable while remaining capable of change. CRR names three operations in that process: Coherence, Rupture, and Regeneration. The public formulation states that systems accumulate historical constraint, close a regime when a bounded condition is reached, and reconstruct through a causal weighting of history [1-5].

The proposed relationship to the Free Energy Principle is complementary. FEP and active inference describe the statistical organisation of self-evidencing systems and the inferential dynamics by which internal states come to parameterise beliefs about external states [10-12]. CRR proposes an endogenous event-time layered over those dynamics: not merely what a system updates toward, but when a current regime should be considered complete and how its history participates in what follows [4,5]. This should be read as a candidate processual clock, not as an established omission or theorem of FEP.

The conceptual compression is therefore simple:

history accumulates → a finite regime closes → history is taken up again

Wall-clock time measures elapsed duration. A CRR clock instead attempts to measure accumulated distinguishable change. Its "hand" is coherence C ; its regime boundary is rupture δ ; its continuity is regeneration R . This creates a natural bridge between cyclic process and historical irreversibility: finite systems can return locally while the record of completed cycles forms a line of history.

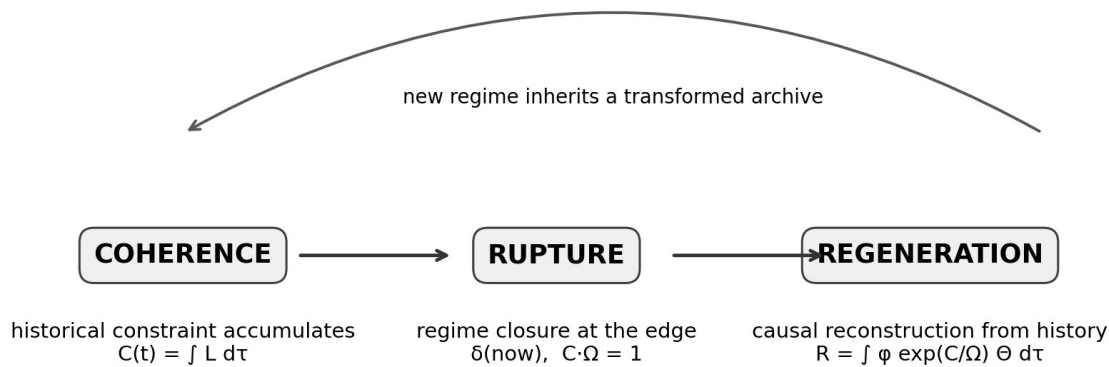


Figure 1. CRR as a minimal temporal grammar: accumulation, closure, and causal reconstruction.

2. Canonical CRR formalism and mathematical glossary

Canonical-source convention

The current public website presents the three equations below and identifies L with Fisher-Rao speed. The current "Final Canonical Formulation" repository instead writes L as the Fisher-Rao squared infinitesimal speed, so that C is a squared-speed action. The main text below reproduces the repository convention while flagging this difference; Appendix G recommends how to resolve it in a full paper [2-4].

2.1 Statistical arena

Let a system at a chosen descriptive scale be represented by a probability distribution $p(\cdot|\theta)$ on a statistical manifold M . The Fisher information matrix supplies the Riemannian metric $G(\theta)$:

$$I_{ij}(\theta) = E_{x \sim p(\cdot|\theta)} [\partial_i \log p(x|\theta) \cdot \partial_j \log p(x|\theta)], \quad G(\theta) = I(\theta) \quad (1)$$

The corresponding infinitesimal Fisher-Rao line element is $ds^2 = d\theta^T G(\theta) d\theta$. Fisher geometry is attractive here because it is invariant under smooth reparameterisation and supplies an intrinsic notion of distinguishability on probability distributions [6-8].

2.2 Coherence: accumulated informational activity

$$C(x,t) = \int_0^t L(x,\tau) d\tau \quad (2)$$

In the current repository convention,

$$L(x,\tau) = (d\theta/d\tau)^T G(\theta) (d\theta/d\tau) = (ds/d\tau)^2 \quad (3)$$

so that

$$C(x,t) = \int_0^t (ds/d\tau)^2 d\tau. \quad (4)$$

Thus C is a non-decreasing historical accumulator when $L \geq 0$. It is intended to encode how much informational work or distinguishable change the current regime has accumulated. The 132-system paper and the original v01 paper instead used the unsquared Fisher-Rao speed, giving arc length rather than action; that iteration is documented in Appendices A-B [1,4].

2.3 Rupture: the present as regime closure

$$\delta(\text{now}) \quad \text{when} \quad C \cdot \Omega = 1 \quad (5)$$

The Dirac delta δ marks an idealised event of zero temporal duration and unit mass. CRR interprets it as the closure of one regime and the opening of another. The equality $C \cdot \Omega = 1$ is a CRR modelling postulate linked in the public materials to Cramér-Rao saturation; the underlying Cramér-Rao inequality itself is

$$\text{Var}(\hat{\theta}) \geq 1 / I(\theta). \quad (6)$$

The step from an efficient-estimation equality to physical or processual rupture is not supplied by estimation theory alone; it is the substantive CRR hypothesis. In the processual reading developed here, rupture is best understood as a boundary-relative completion condition: the current regime has become maximally resolved by its own sufficient statistics, while what follows is not fully determined by those statistics.

2.4 Regeneration: causal reconstruction from history

$$R[\chi](x,t) = \int_{-\infty}^t \varphi(x,\tau) \cdot \exp(C(x,\tau)/\Omega) \cdot \Theta(t-\tau) d\tau \quad (7)$$

The reconstruction resource φ carries available historical material; $\exp(C/\Omega)$ weights material according to recorded coherence; and the Heaviside function Θ enforces causal ordering. The exponential form is structurally related to maximum-entropy exponential families when C is treated as a sufficient statistic and $1/\Omega$ as a natural parameter [3,9]. CRR makes a distinctive commitment here: weighting is by historical coherence, not ordinary elapsed-time recency.

2.5 The Ω parameter and current geometric interpretation

The current canonical repository distinguishes a geometric Ω from an intrinsic rupture precision. For a continual memory-bearing compact manifold with selected closed-geodesic length Φ_G , it writes

$$\Omega_{\text{geo}} = 1 / \Phi_G. \quad (8)$$

The repository then separately introduces Ω_{int} for a normalised intrinsic Z_2 rupture. This disambiguation is necessary because using the same Ω simultaneously as a geometric inverse length, a variance, and a kernel temperature can otherwise create unit and scaling ambiguities [3]. In this document, Φ_G denotes geodesic length and φ denotes the historical resource field.

2.6 Current predictive module

A later CRR iteration adds a parameter-free variability prediction under a canonical Bernoulli(1/2) rupture-noise model:

$$E[\tau] = 1/\Omega, \quad \text{std}(\tau) = 1/2, \quad \text{CV} = \Omega/2. \quad (9)$$

Together with $\Omega_G = 1/\Phi_G$, this gives $\text{CV}_G = 1/(2\Phi_G)$. In the present CRR literature, a bistable Z_2 class is assigned $\Phi = \pi$ and a rotational $\text{SO}(2)$ class $\Phi = 2\pi$, producing $1/(2\pi) \approx 0.159$ and $1/(4\pi) \approx 0.080$ respectively [3,4].

These are empirical predictions of the later geometric/CV programme, not necessary for using CRR as a heuristic processual clock.

2.7 Mathematical glossary

Term	Mathematical / processual meaning
M	Statistical manifold on which the chosen system description lives.
$p(\cdot \theta)$	Probability distribution used to represent the system at the chosen descriptive scale.
θ	Coordinates or parameters of the instantaneous probability distribution.
$G(\theta)$	Fisher information metric; $G = I$ in the canonical statistical geometry.
ds	Infinitesimal Fisher-Rao distance between neighbouring distributions.
x	Spatial, state, or contextual label carried by the field notation; its meaning is domain-specific.
t	Current time of evaluation.
τ	Historical integration variable.
$L(x,\tau)$	Coherence accumulation rate. Current repository: squared Fisher-Rao speed; v01/132-paper: Fisher-Rao speed.
$C(x,t)$	Accumulated coherence: the historical integral of L since the chosen regime origin.
C^*	Coherence value at regime closure. Under $C \cdot \Omega = 1$, $C^* = 1/\Omega$ in the units being used.
$\delta(\text{now})$	Dirac delta marking an idealised rupture or regime-closure event.
$\Theta(t-\tau)$	Heaviside step function enforcing that only settled past contributes to regeneration.
$\varphi(x,\tau)$	Historical resource/archive field: material available to be drawn on during regeneration.
$R[\chi](x,t)$	Regeneration functional: the reconstructed contribution obtained from the weighted causal archive.
Ω	CRR scale/precision symbol. Current materials use both Ω_{geo} and Ω_{int} ; it should not be treated as one unqualified physical quantity.
Φ_G	Notation used in this document for the selected closed-geodesic length of a phase manifold G .
$\exp(C/\Omega)$	Historical weighting kernel; lower effective temperature produces sharper weighting, higher effective temperature flatter weighting, subject to convention.

CV	Coefficient of variation of inter-event intervals: standard deviation divided by mean. A later CRR predictive module proposes $CV = \Omega/2$ under a specific rupture-noise model.
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3. Five core processual axioms

The following axioms are a distilled processual reading of the canonical mathematics. They overlap with the five commitments used in the 132-system paper, but are worded to keep the heuristic CRR core distinct from stronger numerical claims [4].

Axiom 1 - Historical accumulation. A persistent finite system acquires path-dependent structure. At a chosen descriptive scale this can be represented by a non-negative accumulation functional $C(t)$ over its informational trajectory.

Axiom 2 - Finite-regime closure. A particular regime cannot support indefinite distinguishability under a fixed descriptive organisation. Rupture marks the closure of that regime. The exact threshold law is a model choice to be empirically justified; $C \cdot \Omega = 1$ is the current CRR proposal.

Axiom 3 - Causal regeneration. A new regime cannot be reconstructed from an unrealised future. Its available material is the settled causal past plus current arrivals across its boundary. Regeneration therefore has a causal kernel $\Theta(t-\tau)$ and a historically structured resource field ϕ .

Axiom 4 - Edge complementarity. Immediately before closure, the current regime is maximally resolved relative to its own sufficient statistics, while the next regime is maximally unresolved by those statistics alone. External constraints are therefore not "nothing"; they are information not yet internal to the exhausted regime.

Axiom 5 - Relational inheritance across scales. A completed event for one finite system can become historical material for another. Nested systems therefore inherit and transform one another's completed processes, allowing local cycles to compose larger-scale clocks without requiring one global clock.

Parsimony

These five axioms reduce the heuristic CRR grammar to three temporal verbs - integrate, close, reconstruct - plus two constraints: finitude and causal inheritance. The stronger geometric identifications, CV predictions, and metaphysical interpretations can then be tested or debated as modules rather than being smuggled into the core by definition.

4. Feminine and Masculine as archetypal process structure

Interpretive status

This section uses "masculine" and "feminine" as symbolic or archetypal polarities, not as biological essences or prescriptions for men and women. Several traditions historically gender these polarities, but the CRR interpretation below deliberately makes them relational functions that any finite system can enact. The correspondences are comparative, not derivations.

4.1 The first mapping

A recurring symbolic polarity across analytical psychology and contemplative traditions can be expressed as differentiation/receptivity, form/field, line/circle, emission/reception. Jung used Logos as a conceptual aid for discrimination, judgement, and cognition and Eros for relatedness, while later Jungian work has emphasised that these functions cannot responsibly be reduced to fixed properties of biological sex [14,15]. Chinese yin-yang cosmology is even more explicitly relational: yin and yang are correlative, context-dependent, mutually defining categories rather than substances [16].

CRR function	Archetypal "masculine" pole	Archetypal "feminine" pole
Distinction	discrimination, boundary, definition	relation, context, field

Geometry	line, vector, axis	circle, enclosure, matrix
Knowing	"this, not that"	holding ambiguity
Action	projection, emission, giving form	reception, incorporation, transformation
Time	direction, count, biography	recurrence, rhythm, return
Structure	form	material/field from which form arises
Rupture	cut, separation, commitment	surrender, dissolution, opening
Regeneration	new determination	gestation, recombination, birth

CRR immediately destabilises any static reading of this table. Coherence requires differentiation inside a containing field. Rupture is both a cut and a surrender of the existing form. Regeneration is gestation followed by a new distinction. The more faithful image is therefore not two fixed columns but a reciprocal transformation of poles.

4.2 Jung, Logos/Eros, and the coniunctio

In Jungian language, Logos differentiates: this is this, and that is not this. Eros relates: this exists in connection with that. Read through CRR, Logos resembles the progressive articulation of a regime, while Eros resembles the relational field through which exhausted form can receive what it cannot derive from itself. The point is not that men are Logos and women Eros; Jung himself described the concepts as schematic aids, and contemporary scholarship treats the historical gender coding critically [14,15].

The alchemical coniunctio offers a further resonance. Jung interpreted alchemical king/queen and Sol/Luna imagery as symbolic of psychic opposites undergoing dissolution and recombination. CRR gives this ancient pattern a temporal grammar:

form → dissolution / rupture → re-formation

This is not evidence that alchemy anticipated CRR. It is evidence that the experiential pattern "form becomes exhausted, dissolves, and returns otherwise" has repeatedly attracted symbolic representation.

4.3 Circle and line: recurrence married to history

Symbolically, circles are often coded lunar, receptive, or feminine, while lines and axes are coded solar, directional, or masculine. CRR makes neither sufficient. Pure circularity returns but has no biography. Pure linearity proceeds but has no regenerative return. If a cyclic phase $\theta \in S^1$ is paired with a rupture count n , the lifted coordinate $\tilde{\theta} = \theta + 2\pi n$ unwraps the circle into a historical line. Recurrence creates the circle; rupture creates the count; the count creates direction. The "child" of circle and line is historical time.

4.4 Daoism: polarity as reversal rather than identity

Yin-yang is an especially useful comparison because the terms are relational and dynamic. Chinese philosophical sources associate yang with force, brightness, and at times masculinity, and yin with yielding, darkness, and at times femininity; yet the two exist only in relation and processes are frequently understood as cycling, reaching extremes, and reversing [16].

At the CRR edge, the current regime is as internally articulated as it can become. Archetypally this resembles maximum yang: maximum form, maximum distinction. Precisely there, however, the current form can no longer derive its own nextness. It becomes receptive to what lies beyond its sufficient statistics. Maximum determination turns into openness; yang turns toward yin. Regeneration then receives, reorganises, and progressively differentiates again.

4.5 The relational weave: the same event changes polarity across a boundary

A central CRR intuition is that what leaves one system as an event can arrive in another as historical material. In v01 this is stated in the form "what is ϕ for me is δ from you" [1]. Archetypally, an event is active or "masculine" relative

to the system that emits or completes it, yet receptive or "feminine" relative to the system that receives and transforms it. The polarity belongs to the direction of relation, not to the substance of the event.

$$\delta_A \rightarrow \varphi_B \quad \text{and later} \quad \delta_B \rightarrow \varphi_A$$

This is close to Ibn 'Arabī's use of gendered metaphors as universal principles of activity and receptivity: the same being can be active in one relation and receptive in another, and these principles are applied cosmically rather than only to human sex [17]. CRR therefore makes the polarity reversible by construction.

4.6 Tantra and Vajrayāna: the feminine is not passive

Tantric traditions provide an important correction to a simplistic active-masculine/passive-feminine binary. The Oxford Handbook of Tantric Studies emphasises transformation, embodiment, gendered cosmic power, goddess traditions, and dynamic energies; in nondual Śaiva traditions Śakti is not inert matter but the dynamic power of manifestation [18]. A CRR archive or holding field is likewise not passive: what is received is transformed into the conditions of new form.

Vajrayāna traditions often symbolically pair method or skillful means (upāya) with wisdom or insight (prajñā). Sarah Jacoby's work on Sera Khandro describes the tantric ideal as an inseparable union of method and insight rather than one pole simply dominating the other [19]. CRR resonates structurally with this refusal of one-sidedness: precision without openness becomes reification; openness without form cannot stabilise a world.

$$\text{precision without reification} + \text{openness without dissolution}$$

4.7 Kabbalah, holding, and the created space

Kabbalistic traditions developed complex gendered symbolism around masculine and feminine powers within the divine realm, including traditions that place substantial emphasis on their union and on the irreducibility of the feminine [20]. CRR v01 also compares Lurianic contraction to Winnicottian graduated withdrawal: creation requires a holder who does not fill all available possibility [1]. Too much determination leaves no room for the other to become; too little holding leaves no stable field from which regeneration can proceed.

4.8 The poised edge and the contemplative pivot

The deepest archetypal reading appears at the edge. The system is maximally itself relative to the distinctions of the current regime: maximum Logos, maximum form, maximum boundary. And precisely there it is least able to derive the next distinction from itself. The spear-tip becomes the opening. The line reaches its end and encounters the circle. A receiving field takes up what the old form could not contain, and a new differentiation begins.

Healthy archetypal functioning therefore cannot mean choosing one pole. A healthy "masculine" function is discernment without domination, boundary without severance, commitment without closure. A healthy "feminine" function is receptivity without engulfment, holding without possession, openness capable of giving birth to form. The contemplative pivot is the capacity to move between these orientations without identifying absolutely with either.

This is also why Zhuangzi is suggestive for CRR. The original v01 paper invokes the "pivot" as an image of responsiveness that is possible because the centre is not fixed to one determination [1]. The CRR now can therefore be read as a hinge: determinate past | present conversion | undetermined becoming.

5. Platonic ingression: a CRR extension over potential forms

Proposed extension

The mathematics in this section is not part of the present canonical CRR equations. It is a constructive extension designed to ask whether CRR can represent Whiteheadian/Platonic ingression, multiscale agency, and history-conditioned choice without adding a second metaphysical force. The extension is mathematically explicit so that its assumptions can be inspected independently.

5.1 Whiteheadian motivation: actuality as decision amid potentiality

Whitehead replaces enduring substance with actual occasions: finite processes of concrescence that inherit an antecedent world, realise some potentials, and become objective data for later occasions. He calls eternal objects pure potentials and describes actuality as decision amid potentiality; objective eternal objects include mathematical Platonic forms [21]. This provides a precise philosophical target for a CRR extension: can the historical archive be resolved by candidate forms, so that regeneration becomes a changing valuation over possibilities?

5.2 Index the archive by candidate form

Let a system i at scale s have a set of realizable forms $E_s = \{e_1, e_2, \dots\}$. A form can mean a policy, morphology, interpretation, behavioural configuration, or other candidate pattern at the chosen scale. Replace the undifferentiated archive $\varphi(t)$ by $\varphi_e(t)$, the historical material relevant to candidate e . Then define

$$R_e(t) = \int_{-\infty}^t \varphi_e(\tau) \cdot \exp(C(\tau)/\Omega) d\tau. \quad (10)$$

Normalising across candidate forms yields a historical prior:

$$P_{\text{hist}}(e|t) = R_e(t) / \sum_f R_f(t). \quad (11)$$

The eternal or possible form is not created by history in this reading. History changes its relevance to this becoming. That is close to Whitehead's language of "valuation up" and "valuation down" of eternal objects [21].

5.3 Couple the historical prior to active inference

Active inference already supplies a prospective weighting over policies or candidate actions in terms of expected free energy. To avoid confusing the Fisher metric G with expected free energy, write the latter as $\mathcal{G}(e,t)$. A natural coupling is

$$Q(e|t) = [P_{\text{hist}}(e|t) \cdot \exp(-\gamma \mathcal{G}(e,t))] / \sum_f [P_{\text{hist}}(f|t) \cdot \exp(-\gamma \mathcal{G}(f,t))]. \quad (12)$$

This produces a parsimonious division of labour. CRR supplies an evolving historical prior; active inference supplies prospective valuation of viable futures [10-12]. Becoming is then the product of historical inheritance and future viability, without introducing a separate optimisation principle.

$$\text{becoming} \propto \text{historical inheritance} \times \text{prospective viability}$$

5.4 Choice is prepared continuously, not created ex nihilo at the cut

Because $dR_e/dt = \varphi_e(t) \exp(C/\Omega)$, the normalised historical probabilities $p_e = R_e/\sum_f R_f$ obey

$$dp_e/dt = [\exp(C/\Omega) / \sum_f R_f] \cdot [\varphi_e - p_e \sum_f \varphi_f]. \quad (13)$$

The potential forms are therefore being re-valued throughout coherence. The cut need not be imagined as a magical point at which an arbitrary choice suddenly appears. It is the closure of an ongoing concrescence: continuous valuation $\rightarrow$ poised edge $\rightarrow$ actualisation.

5.5 The beauty/poise function

The current CRR website proposes a "beauty function"

$$B(C) = \exp(C/\Omega) \cdot (C^* - C). \quad (14)$$

Differentiation gives $dB/dC = \exp(C/\Omega)[(C^* - C)/\Omega - 1]$, so the maximum occurs at

$$C = C^* - \Omega. \quad (15)$$

This is a useful heuristic for agency. Too early, there is remaining possibility but insufficient historical determination. At rupture, there is determination but no remaining room inside the regime. One Ω -unit before closure, historical weight and residual capacity are jointly high. The public site interprets this as the poised edge rather than the rupture itself [2].

5.6 Outside uncertainty as boundary-relative information

Let X denote everything represented by the exhausted regime, Y the arriving causal past of coupled systems, and Z the next realised form. The relevant information-theoretic statement is not that the external world has objectively maximal entropy. It is that the next state can remain unresolved conditional on X while becoming more constrained when Y is available:

$$H(Z|X,Y) \leq H(Z|X). \quad (16)$$

This formalises a boundary-relative version of the claim: my current regime cannot fully determine my future, but the causal past outside me can constrain it. "Outside" is therefore not blankness; it is information not yet internal to the exhausted regime.

5.7 I witness the past as my future

Let $N_j(t)$ count rupture events completed by another finite system j , and let $K_{ji}(t-\tau)$ be a causal coupling kernel carrying their effects to system i . A minimal relational archive is

$$\varphi_i(t) = \varphi_i^{\text{self}}(t) + \sum_j \int_{-\infty}^t K_{ji}(t-\tau) dN_j(\tau). \quad (17)$$

The chain is then:

their actuality $\rightarrow$ my archive $\rightarrow$ my possibility weighting $\rightarrow$ my actuality $\rightarrow$ their archive

Signals always arrive from a causal past. A finite system therefore does not encounter another system's inaccessible interior "now"; it encounters consequences that have reached its boundary. This resonates with Whitehead's causal past and with CRR v01's reciprocal δ/ϕ relation [1,21]. It also fits multiscale biological perspectives in which agents are nested within other agents and have different cognitive boundaries or "light cones" of states they can represent and act upon [23,24].

5.8 Circle, line, and tree

A cyclic phase $\theta \in S^1$ becomes a historical line when completed cycles are counted: $\tilde{\theta} = \theta + 2\pi n$. Add more than one candidate continuation at each rupture and a third geometry appears. The circle is recurrence; the line is the realised sequence; the tree is the unrealised branching of possible histories.

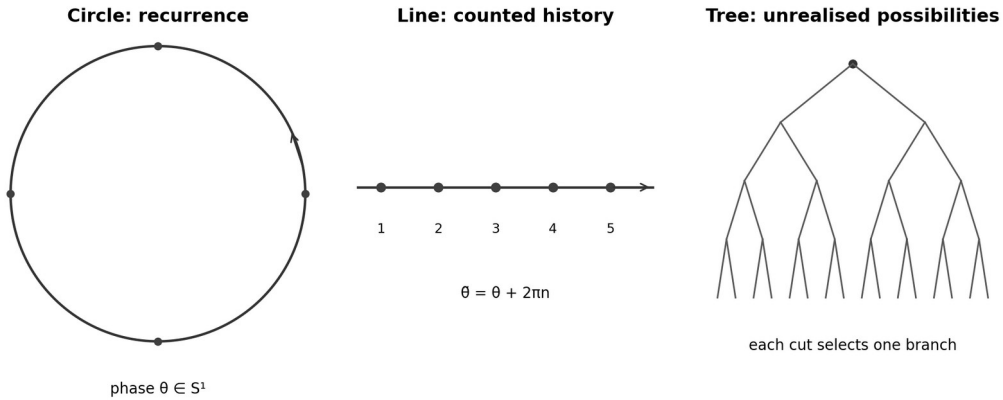


Figure 2. A proposed process geometry for ingress: recurrence (circle), realised history (line), and unrealised potentiality (tree).

If each cut has two possible continuations, N cuts generate 2^N possible branches before constraints are imposed. CRR's archive and FEP weighting together define a measure over this tree rather than requiring every branch to become actual.

5.9 A path measure over becoming

For a realised history $\Gamma = (e_1, \dots, e_N)$, let the probability of each selected form be the active-inference coupling in Equation 12. Then

$$P[\Gamma] = \prod_n Q_n(e_n) = \exp(-S[\Gamma]), \quad (18)$$

with an effective path action

$$S[\Gamma] = \sum_n [\gamma \mathcal{G}_n(e_n) - \log P_{\text{hist},n}(e_n) + \log Z_n]. \quad (19)$$

A path is favoured when its prospective expected free energy is low and its forms are historically supported. This is not a derivation of Whitehead, but it gives "ingression" a transparent stochastic meaning: history-conditioned realisation of form.

5.10 Historical choice as reinforcement

A toy model makes the consequence visible. Suppose two forms A and B begin with equal archive mass W, and each actualisation reinforces the selected form by an edge weight $w = \exp(1/\Omega^2)$ under $C^*\Omega = 1$. This creates a Pólya-like reinforced process [27]. If $a = W/w$, the symmetric limiting share p of A has a Beta(a,a)-like concentration; one simple concentration statistic is $E[p^2 + (1-p)^2] = (a+1)/(2a+1)$. For $W = 100$:

Ω	edge reinforcement $\exp(1/\Omega^2)$	expected concentration
1.0000	2.7183	0.5067
0.7000	7.6969	0.5185
0.5000	54.5982	0.6072
0.3183	19,333.6891	0.9949

The point is not the literal numerical fit of this toy urn. It is the structural consequence: if completed choices are re-entered into the archive, historical actualisation can progressively reshape the probability of future ingression. The very large canonical edge weight also warns that any empirical version must normalise archive mass and kernel units carefully.

5.11 Dense actualisation across scales, and the finitude constraint

Let scale s have characteristic period $T_s = T_0/b^s$ for $b \geq 2$. The union of rupture-time grids $D_s = \{nT_0/b^s\}$ is dense in time: for any non-empty interval, a sufficiently fine scale has spacing smaller than that interval, so some rupture lies inside it. Thus "choices at every moment across scales" can be made precise as dense event structure without requiring every real-valued instant to be a rupture for any one system.

But an infinite hierarchy of independent one-bit ruptures would have information rate

$$\bar{I}_S = \sum_{s=0}^S 1/T_s = [b^{S+1}-1] / [(b-1)T_0], \quad (20)$$

which diverges as $S \rightarrow \infty$. A finite world therefore requires a smallest effective scale, declining entropy per event, or cross-scale correlation/compression. This is where multiscale agency becomes important. Levin's TAME framework explicitly treats organisms as nested agents with goals at different scales, and Watson and Levin argue that collective and individual intelligence are not categorically separate [23,24]. CRR can use such nesting without assuming statistically independent "choices" at every scale.

5.12 Platonic forms as cross-scale invariants

Suppose a coarse-graining operator $\mathcal{R}$ maps lower-scale patterns to higher-scale descriptions. An eigenform F_α satisfies

$$\mathcal{R}F_\alpha = \lambda_\alpha F_\alpha. \quad (21)$$

If $\lambda_\alpha = 1$, then $\mathcal{R}F_\alpha = F_\alpha$: the pattern is invariant under the chosen scale transformation. Renormalisation-group theory uses fixed points and eigen-directions to explain how large-scale regularities can become insensitive to microscopic detail [25,26]. This provides a mathematically disciplined analogue of a "Form": not a ghostly object outside spacetime, but a pattern that survives coarse-graining.

Expand the archive as $\varphi_s(t) = \sum_\alpha a_{\alpha,s}(t)F_\alpha$. History changes the coefficients a_α - the valuation or prevalence of the form - while the invariant mode supplies a recurring structural possibility. On a Whiteheadian reading, the form is potential and choices change its degree of ingression. On a more naturalistic reading, the stable form may itself emerge as an attractor or fixed point of repeated relational dynamics. CRR does not decide between these metaphysical interpretations.

5.13 Archetypes as cross-scale attractors: a final speculative bridge

If recurring relational problems - self/other, union/separation, parent/child, approach/withdrawal, destruction/repair - repeatedly project multiscale systems onto a small family of stable modes, then a Jungian archetype could be interpreted as a cross-scale attractor in the historical possibility field. This would explain how concrete manifestations vary while recognisable relational structure recurs. It is a speculative bridge, not an empirical identification of Jungian archetypes with renormalisation fixed points.

6. Conclusion

CRR is most parsimonious when treated first as a grammar of finite becoming. A system accumulates historically structured distinction; a regime closes; what follows is rebuilt from causal history. The Free Energy Principle can supply within-regime inferential dynamics, while CRR proposes an endogenous process clock that marks when accumulated change becomes a new regime. The core insight is therefore less "everything ruptures at a universal number" than "persistence requires a way to turn completed distinction into historical constraint and to take that history up again."

This process reading also clarifies circle and line. Locally, finite systems recur. Globally, completed recurrence is counted and becomes irreversible history. The archive is not simply memory storage: it is the structured causal past available to present becoming. At the edge, the system may be maximally resolved relative to its own regime while the next state remains underdetermined by that regime alone. What arrives from outside is therefore not an ahistorical external present but the causal past of other finite systems reaching this boundary.

The archetypal comparison follows naturally. Masculine and feminine become alternating relational orientations rather than fixed kinds of person: giving form and receiving, discriminating and holding, line and circle. Every finite system can enact both depending on which side of a boundary and which phase of a process is being described. The contemplative "pivot" is not either pole but the capacity for polarity to reverse without continuity being destroyed.

The Platonic-ingression extension then takes one further step. By indexing the archive over possible forms, normalising it into a historical prior, and coupling that prior to expected free energy, CRR becomes a history-dependent measure over potential futures. Completed acts become constraints for other systems; nested clocks make actualisation dense across scales; coarse-graining suggests a disciplined way to speak of recurrent forms without claiming that mathematics has proven Platonic metaphysics. Whiteheadian ingression, active inference, Levin's multiscale agency, and renormalisation theory meet here as structurally compatible languages rather than as one derived theory.

The strongest route forward is therefore modular. Preserve the minimal temporal grammar. Make the Fisher-geometric clock explicit and dimensionally clean. Treat rupture thresholds, Ω -identifications, CV baselines, and Lie-group predictions as falsifiable modules rather than definitions. Keep archetypal and metaphysical correspondences clearly labelled as comparative interpretations. On those terms CRR can go very far: not because three equations explain everything, but because three temporal operations may provide a compact syntax in which very different finite processes can be compared without pretending that their mechanisms are the same.

References

- [1] Sabine, A. (2026). The Paradox That Must Not Be Resolved: Transitional Space, the Relational Weave, and a Temporal Grammar for Finite Systems. Exploratory working paper, version 01, August 2026.
- [2] Sabine, A. (2026). CRR: A Temporal Grammar. temporalgrammar.ai / cohere.org.uk. Public web formulation, accessed 30 August 2026.
- [3] Sabine, A. (2026). CRR - Final Canonical Formulation. CRR GitHub repository, main branch. Accessed 30 August 2026.
- [4] Sabine, A. (2026). Coherence-Rupture-Regeneration: a geometric framework for temporal variability tested across neural oscillations and 132 oscillatory systems. Preprint.
- [5] Sabine, A. (2026). Commentary: Differential Flatness and CRR. cohere.org.uk, January 2026.
- [6] Rao, C. R. (1945). Information and the accuracy attainable in the estimation of statistical parameters. *Bulletin of the Calcutta Mathematical Society*, 37, 81-91.
- [7] Čencov, N. N. (1982). *Statistical Decision Rules and Optimal Inference*. American Mathematical Society.
- [8] Amari, S. (2016). *Information Geometry and Its Applications*. Springer.
- [9] Jaynes, E. T. (1957). Information theory and statistical mechanics. *Physical Review*, 106, 620-630.
- [10] Parr, T., Da Costa, L., & Friston, K. (2020). Markov blankets, information geometry and stochastic thermodynamics. *Philosophical Transactions of the Royal Society A*, 378, 20190159.

- [11] Kirchhoff, M., Parr, T., Palacios, E., Friston, K., & Kiverstein, J. (2018). The Markov blankets of life: autonomy, active inference and the free energy principle. *Journal of the Royal Society Interface*, 15, 20170792.
- [12] Friston, K. (2019). A free energy principle for a particular physics. arXiv:1906.10184.
- [13] Winnicott, D. W. (1971). *Playing and Reality*. Tavistock.
- [14] Jung, C. G. (1959/1968). *Aion: Researches into the Phenomenology of the Self*. Collected Works, Vol. 9, Part II. Princeton University Press.
- [15] Bowles, M. L. (1993). Logos and Eros: The vital syzygy for understanding human relations and organizational action. *Human Relations*, 46(11), 1271-1290.
- [16] Wang, R. R. (2012). *Yinyang: The Way of Heaven and Earth in Chinese Thought and Culture*. Cambridge University Press. See also Stanford Encyclopedia of Philosophy, "Gender in Confucian Philosophy" (2023).
- [17] Shaikh, S. (2012). *Sufi Narratives of Intimacy: Ibn 'Arabi, Gender, and Sexuality*. University of North Carolina Press.
- [18] Payne, R. K., & Hayes, G. A. (eds.) (2024). *The Oxford Handbook of Tantric Studies*. Oxford University Press.
- [19] Jacoby, S. H. (2014). *Love and Liberation: Autobiographical Writings of the Tibetan Buddhist Visionary Sera Khandro*. Columbia University Press.
- [20] Idel, M. (2024). On Equality and Irreducibility of Female to Male in Safedian Forms of Kabbalah. *Zutot*, 21(1).
- [21] Whitehead, A. N. (1929/1978). *Process and Reality*, corrected ed. Free Press. See also the Stanford Encyclopedia of Philosophy, "Alfred North Whitehead".
- [22] Simmer-Brown, J. (2023). The *Ḍākinī* in Indo-Tibetan Buddhism. In *The Oxford Handbook of Tantric Studies*. Oxford University Press.
- [23] Levin, M. (2022). Technological Approach to Mind Everywhere: An experimentally-grounded framework for understanding diverse bodies and minds. *Frontiers in Systems Neuroscience*, 16, 768201.
- [24] Watson, R., & Levin, M. (2023). The collective intelligence of evolution and development. *Collective Intelligence*, 2(2).
- [25] Wilson, K. G. (1971). Renormalization Group and Critical Phenomena. I. Renormalization Group and the Kadanoff Scaling Picture. *Physical Review B*, 4, 3174.
- [26] Wilson, K. G. (1975). The renormalization group: Critical phenomena and the Kondo problem. *Reviews of Modern Physics*, 47, 773.
- [27] Pemantle, R. (2007). A survey of random processes with reinforcement. *Probability Surveys*, 4, 1-79.

Appendix A. Iteration 1 - CRR v01 (August 2026)

The uploaded exploratory working paper is the earliest full formalism encountered in this discussion [1]. It defines coherence as Fisher-Rao arc length counted in resolution cells, $C = s/\varepsilon$, with $\varepsilon = 1/\sqrt{N}$ and $\Omega = \varepsilon/\pi$. Rupture occurs when $C\Omega = 1$, equivalent to a traversed arc length π in the chosen binary arena. Regeneration is the historical integral with $\exp(C/\Omega)$, and the paper keeps cycle rate π/L separate from Ω . Its psychological claims depend strongly on the archive retaining coherence as counted in the regime in which an event was lived.

Strengths: clear distinction between clock, archive, kernel, and cut; explicit limitations; useful holding/dyad model; arc-length geometry is easy to interpret. Main vulnerabilities found in stress-testing: accumulated path length can exceed a manifold diameter without physical rupture; stochastic sample paths require a coarse-graining rule; rupture depends on descriptive scale; the universal one-bit Landauer claim is stronger than a generic binary alphabet warrants.

Appendix B. Iteration 2 - Geometric/CV CRR across 132 systems

The later cross-domain paper narrows CRR toward oscillatory variability [4]. It retains C as Fisher-Rao arc length and sets a geometric scale $\Omega = 1/\Phi$, with Z_2 assigned $\Phi = \pi$ and rotational $SO(2)$ assigned $\Phi = 2\pi$. Under a Bernoulli(1/2) rupture-noise model it predicts $CV = \Omega/2$, yielding approximately 0.159 and 0.080. This iteration is more falsifiable because it publishes numerical targets and a classification protocol.

Strengths: explicit assumptions, pre-registration orientation, numerical falsifiers, clearer scientific target. Open issues: the effective-sample-size $n = 1$ step is an additional statistical assumption; topology alone does not fix metric length without normalisation; cross-domain rows are not all independent; the three-class directionality partially reflects how classes are defined. The empirical CV module is promising but should remain separable from the core temporal grammar until independently replicated.

Appendix C. Iteration 3 - Public temporalgrammar.ai / FEP processual CRR

The public site presents CRR as "three equations, one parameter" and explicitly frames it as a temporal grammar for FEP [2]. It foregrounds coherence as cumulative adaptive work, rupture as the present boundary at $C\Omega = 1$, and regeneration as exponentially weighted history. It also introduces a beauty/poise function $B(C) = \exp(C/\Omega)(C^* - C)$, phase-gating interpretations, $Z_2/SO(2)$ channels, and numerous interactive applications.

Strengths: conceptual clarity, strong processual narrative, direct link to FEP, visible simulations. Risks: the site compresses several distinct meanings of Ω - variance, inverse geodesic scale, temperature/precision - and occasionally presents correspondences as more forced than their underlying mathematics warrants. It is an excellent heuristic interface but should not be the sole mathematical specification for a full paper.

Appendix D. Iteration 4 - Final Canonical repository / Lie-group CRR

The current repository attempts to consolidate conventions and extends the geometry to compact connected Lie groups [3]. It distinguishes Ω_{geo} from Ω_{int} , derives $CV_G = 1/(2\Phi_G)$ under its rupture-noise model, and introduces composition and group-specific predictions. It also changes the canonical identification of L to the squared Fisher-Rao speed, making C a squared-speed action rather than arc length.

Strengths: explicit epistemic tiering, convention audit, preregistration discipline, modular claims, Lie-group generalisation. Central mathematical tension: if $C = \int (ds/dt)^2 dt$ while $\Omega_{\text{geo}} = 1/\Phi_G$, then closure after one geometric circuit depends on traversal speed unless unit-speed parametrisation is imposed. The earlier arc-length formulation is reparametrisation-cleaner for a geometry-based capacity argument. The repository itself is commendably explicit that many claims remain T1/T2 rather than established principles.

Appendix E. Iteration 5 - Heuristic processual-clock CRR developed in this discussion

The discussion then treated CRR less as a universal numerical law and more as a processual clock layered onto FEP. In this reading, FEP supplies within-regime inferential dynamics while CRR supplies an endogenous event coordinate. The circle describes recurrence; the rupture count unwraps recurrence into a line of history; the archive is the settled causal past; and coupled systems inherit one another's completed events. A "shared present" is therefore an achievement of entrainment among finite clocks rather than a primitive global time.

This iteration is deliberately heuristic. Its value is parsimony and conceptual reach. Its scientific burden is to define p , L , regime boundaries, and coarse-graining consistently enough that the same clock can be estimated across systems without changing definitions opportunistically.

Appendix F. Iteration 6 - Platonic-ingression / history-conditioned possibility CRR

The final extension indexes the archive by candidate form, ϕ_e , normalises reconstruction into $P_{\text{hist}}(e|t)$, and couples this historical prior to active-inference prospective weighting. It adds a causal cross-system kernel, a path measure over realised histories, and a multiscale coarse-graining interpretation of recurrent forms. In this variant, "ingression" means history-conditioned realisation of form; the circle is recurrence, the line is realised history, and the tree is potentiality.

This is mathematically coherent as a stochastic extension but metaphysically underdetermined. It can support both a Whiteheadian reading in which forms are antecedent potentials and a naturalistic reading in which stable forms emerge as attractors or renormalisation fixed points. No CRR equation by itself decides between those ontologies.

Appendix G. Recommendation for a full paper

Recommendation: take forward a cleaned processual variant, here called CRR-P, and treat the CV/Lie-group programme as a separately testable module. The strongest full paper would preserve CRR's parsimony while eliminating notation overload and preventing geometric, statistical, and metaphysical claims from depending on one symbol.

Recommended core formalism

1. $C(t) = \int_0^t \|d\theta/d\tau\|_G d\tau$ (Fisher-Rao arc length)
2. $s(t) = C(t) / \Phi$ (dimensionless regime phase)
3. rupture at $s = 1$, or with an empirically specified hazard $h(s)$

$$4. R_e(t) = \int_{-\infty}^t \varphi_e(\tau) K_\beta[s(\tau)] \Theta(t-\tau) d\tau$$

$$5. P_{\text{hist}}(e|t) = R_e / \sum_f R_f, \text{ with optional active-inference coupling } Q(e) \propto P_{\text{hist}}(e) e^{-\gamma \mathcal{G}(e)}$$

Here Φ is a clearly defined regime capacity or selected geodesic extent, and β is a separate kernel sharpness/precision parameter. If later evidence demonstrates that β , Φ , rupture noise, and a symmetry class are locked by one geometric invariant, they can be re-identified. They should not be made identical by notation before that result exists.

Why this variant

- Arc length is invariant to reparameterisation and directly comparable to a geodesic extent. Squared-speed action is useful for dissipation or path costs, but it is a different functional and should not silently replace distance.
- The dimensionless phase $s = C/\Phi$ makes the closure condition transparent and removes unit ambiguity from $C\Omega = 1$.
- Separating β (memory weighting) from Φ (capacity) prevents the same Ω from simultaneously acting as inverse length, variance, temperature, and rupture noise scale. Parsimony should be earned empirically, not created by symbol overloading.
- The $CV = \Omega/2$ and Lie-group predictions can then remain a high-value empirical module with clear assumptions. Independent replication can decide whether the parameters collapse after all.
- The archetypal and Platonic sections should be published, if at all, as clearly labelled interpretive extensions or a companion philosophy paper. Their strength is structural resonance, not evidential confirmation of CRR.

The resulting research programme becomes unusually clear. Paper I can ask whether an arc-length process clock predicts regime closure across a small number of carefully operationalised systems. Paper II can test the CV/symmetry predictions prospectively. A separate philosophical paper can develop relational weave, archetypal polarity, and ingression without asking the mathematical results to bear a metaphysical weight they have not yet earned. This preserves what is most distinctive in CRR - its temporal grammar - while making every stronger claim easier to falsify, refine, or retain.